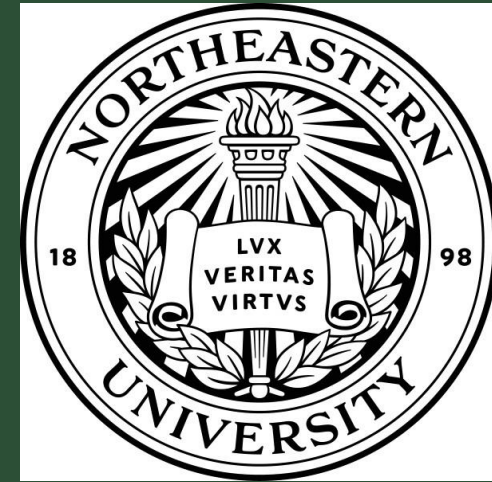


Wildfire Prevention Strategy Using Technology.



Team 1904 Coders.

**Mrityunjay Gupta
Siddharth Alashi
Kush Patel**

The majority of new forest fires started in Canada in recent days are from natural causes, like Lightning. These only Accounts for 67%.

Lightning Strikes have an Outsized Impact, according to Natural Resource Canada.

Figures released by B.C. Wildfire Service(Year 2023) show than more than 460 fires are caused by lightning. Another 275 Were attributed to suspected Human Activities.

The Historical data and Official reports confirms that 59528 in British Columbia, along West Coast. To the east, Alberta had the second highest Number of fire blazes 41657.



Canada Wildfire Analysis



Fire Danger Forecast

Forecast values for the codes and indices can be produced readily from the forecast values of the necessary weather elements.

2 Basic Approaches

Area Averaging



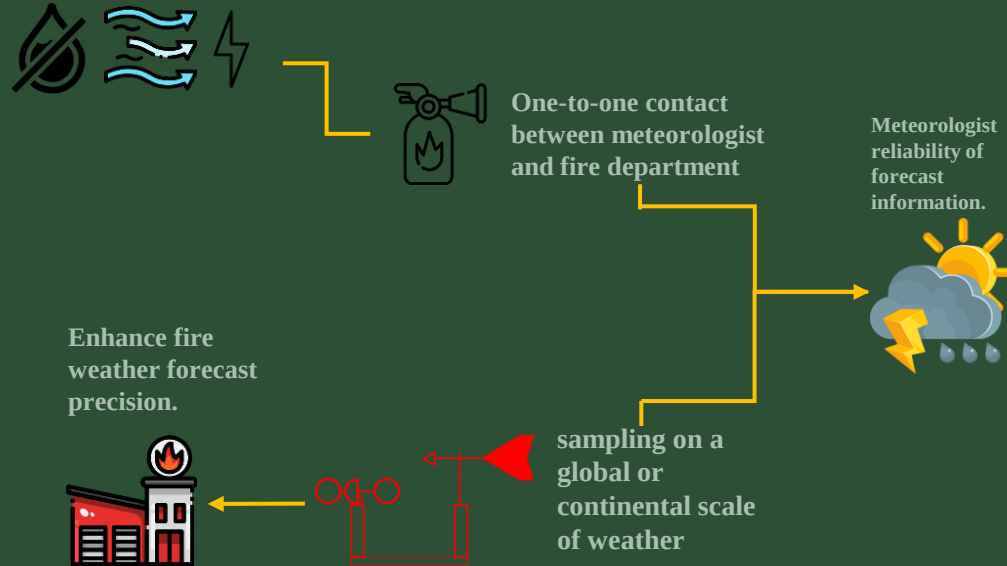
- The observations from fire weather stations are Averaged.
- The forecasted for the next day uses indices of the current day fire danger calculation

Point Forecasting



- Used in mountainous regions
- A number of weather zones and weather observing stations are chosen
- Indices are calculated for the observed and forecasted weather.

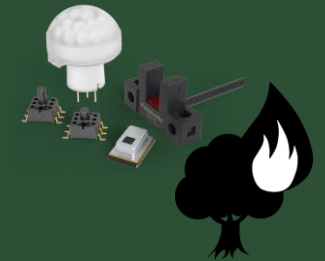
Fire Weather Forecasts



Weather Dataset –
1990 – 2024 (Open
Source)
Wildfire Dataset
(1990-2021)

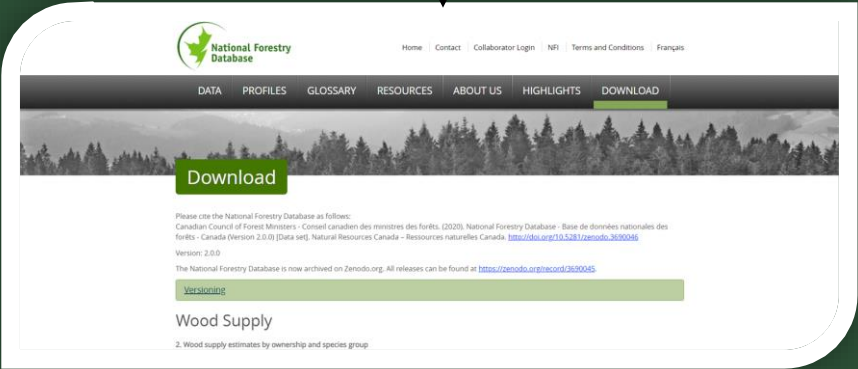
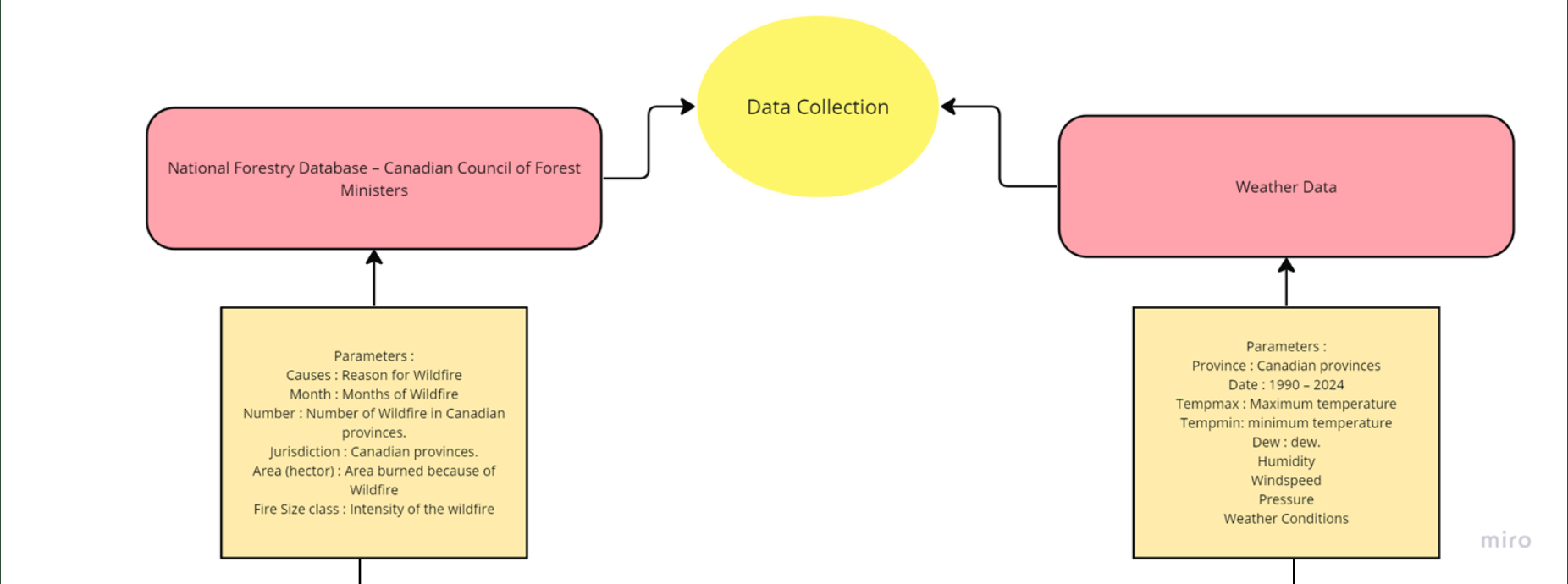


Data Analysis and
Predictive
Modeling



Wildfire Detection
Communication
Sensor

Data Collection Overview:



visualcrossing Weather Data Weather API Query Builder Pricing API Docs More Search docs...

British Columbia Date range: 01/01/1990 - 02/08/2024 Metric: (°C, km)

Addresses, partial addresses or lat,lon

History or forecast data

Query options

Data sections: Weather elements Degree days Wind & solar Agriculture Weather stations

API Grid Chart JSON CSV

Weather Data Additional Data Data Details

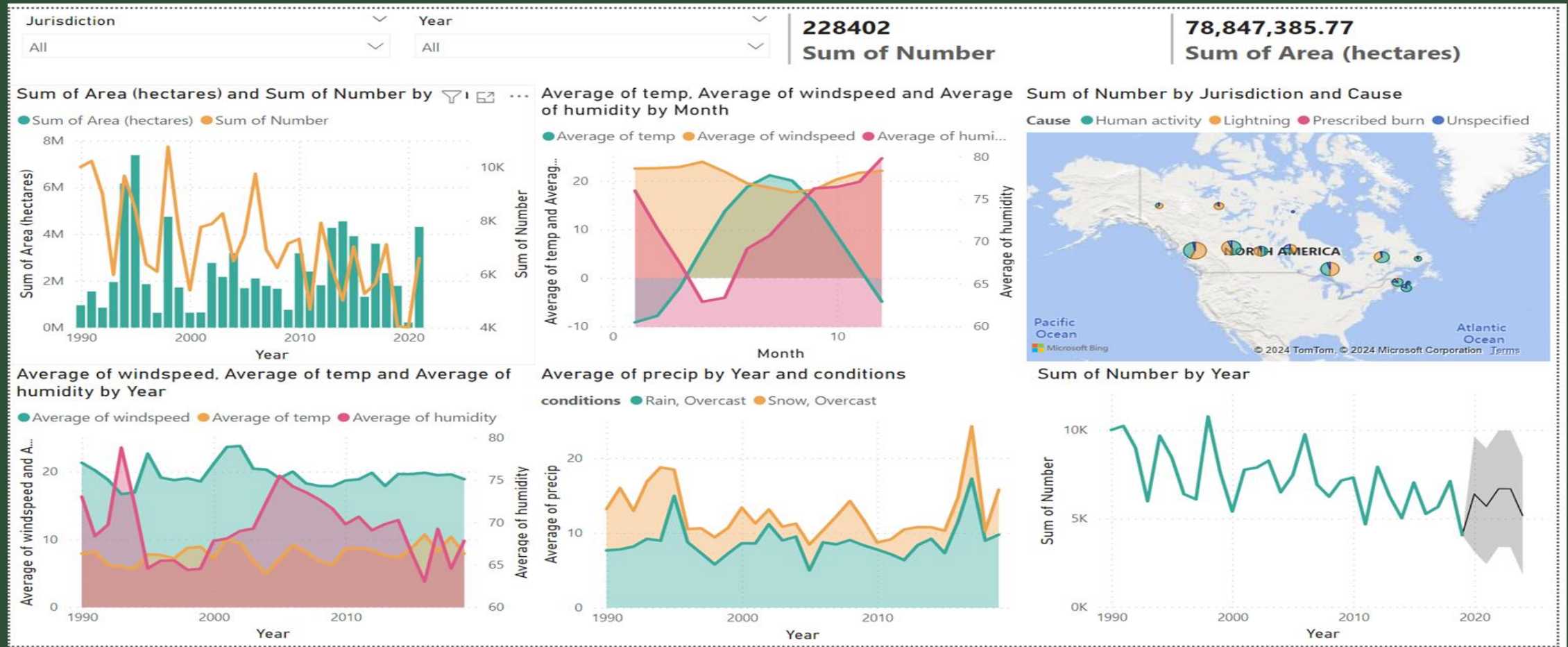
Weather Data **Additional Data** **Data Details**

Weather Data **Additional Data** **Data Details**

Available weather data for BC, Canada. These results are filtered by your query options.

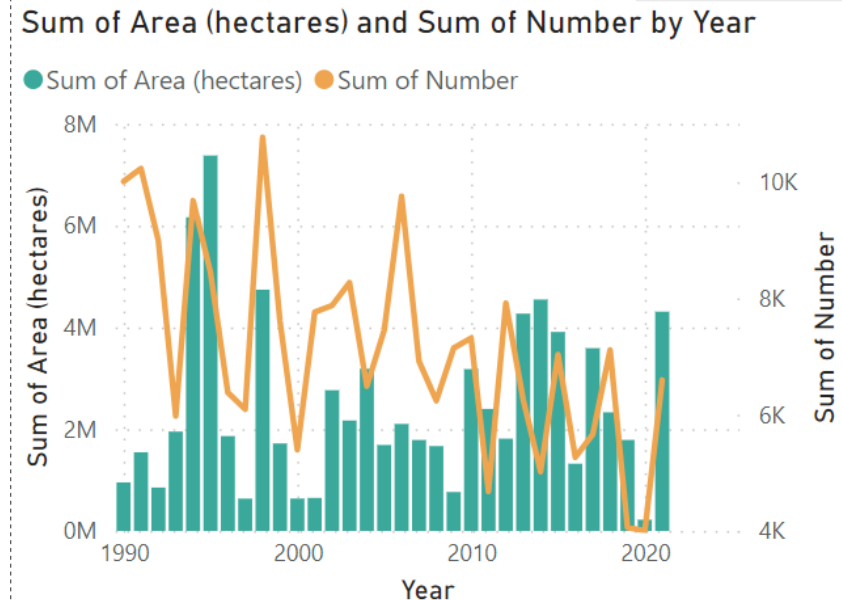
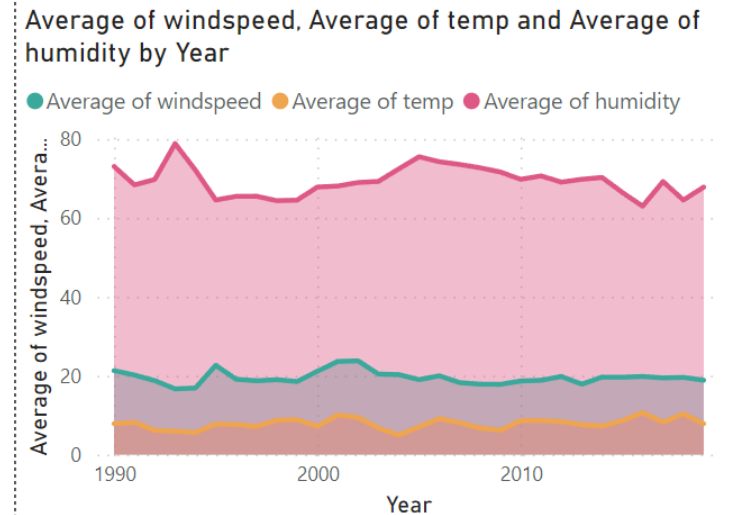
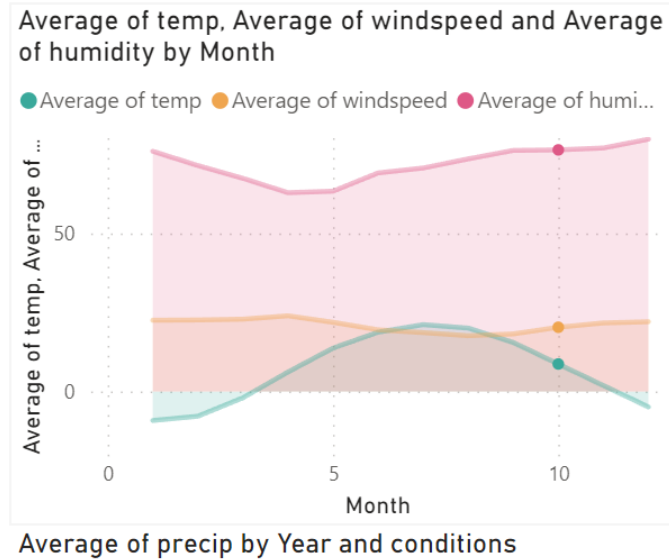
ipcover	precip	type	snow	snowdepth	windgust	windspeed	winddir	pressure	cloudcover	visibility	solarradiation	solarenergy	uvindex	sunrise	sunset	moonphase	conditions
rain	21.9	251.2	1010.2	98.8	30.4									08:05:16	16:29:11	0.17	Rain, Partially cloudy
rain	18.1	249.2	1019.7	78.6	35.4									08:05:12	16:30:11	0.2	Partially cloudy
rain	20.3	176.9	1019.8	94.3	20.5									08:05:06	16:31:13	0.24	Rain, Overcast
rain	22.5	227.6	1017.9	89.4	21.7									08:04:57	16:32:18	0.25	Rain, Partially cloudy
rain	22.6	188.1	1014.8	91.4	17									08:04:45	16:33:25	0.31	Rain, Overcast
rain	28.6	146.3	1007.8	93.1	19.3									08:04:30	16:34:33	0.34	Rain, Overcast
rain	42.3	291.9	999.3	95.9	23.3									08:04:12	16:35:44	0.38	Rain, Overcast
rain														08:04:00	16:36:57	0.42	Rain, Overcast

Dynamic Dashboard



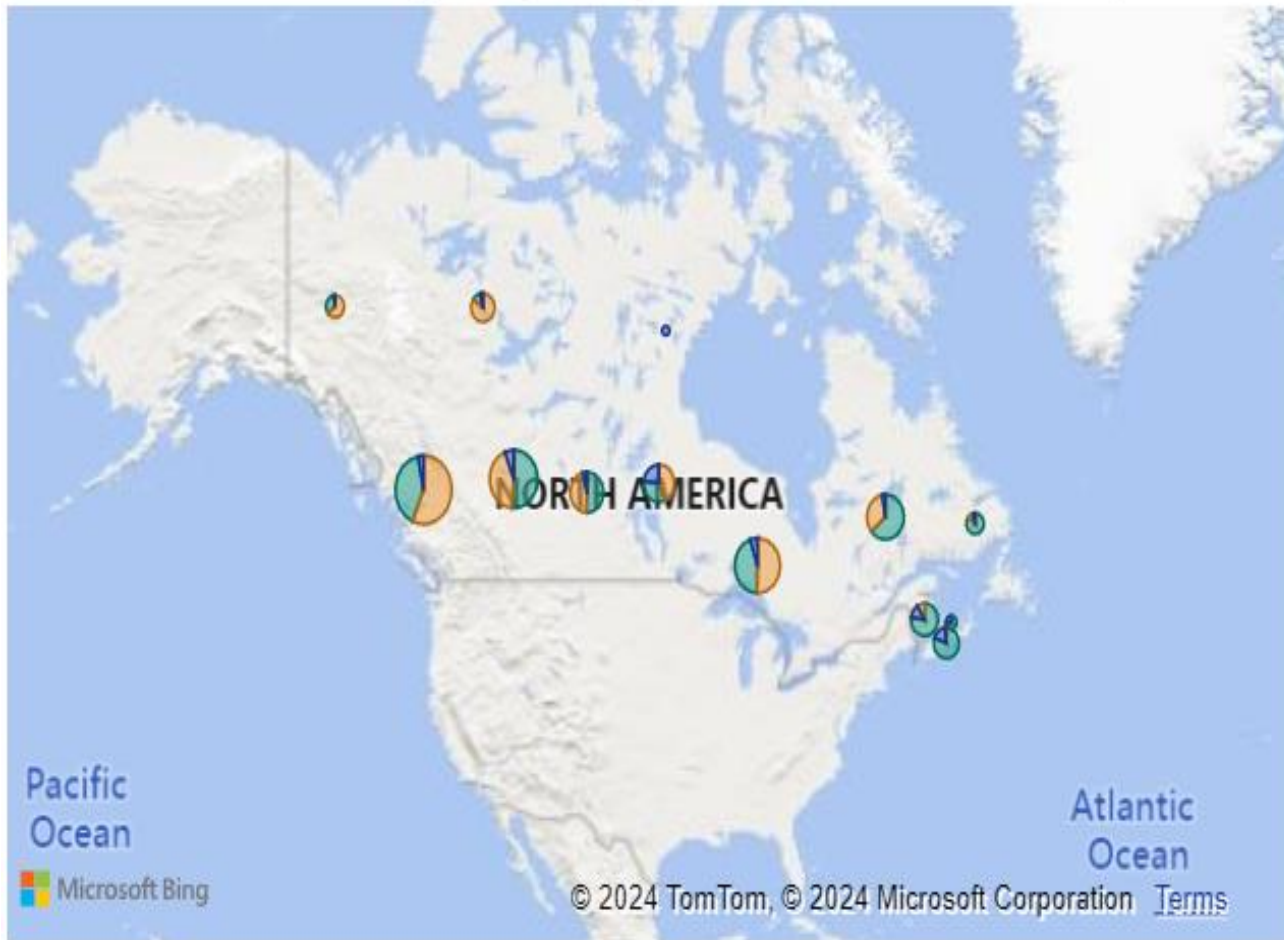
- Dashboard – Dynamic uses filter
- Dropdown : Jurisdiction, Year
- Year : Years 1990 – 2021
- The Notice for the Sum of Area and Wildfires since 1990 -2010
- Predictive Analysis chart for British Columbia

Insights From Dashboard: Area Burnt, Number Of Fires, Temp, Windspeed and Humidity



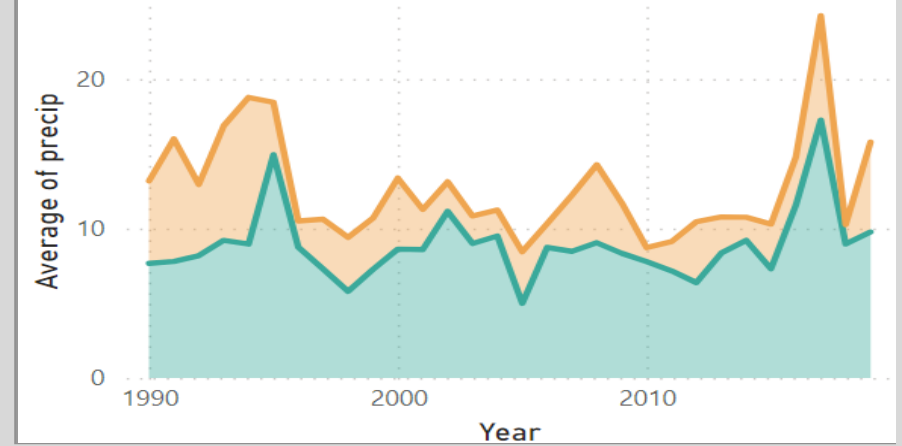
Sum of Number by Jurisdiction and Cause

Cause ● Human activity ● Lightning ● Prescribed burn ● Unspecified

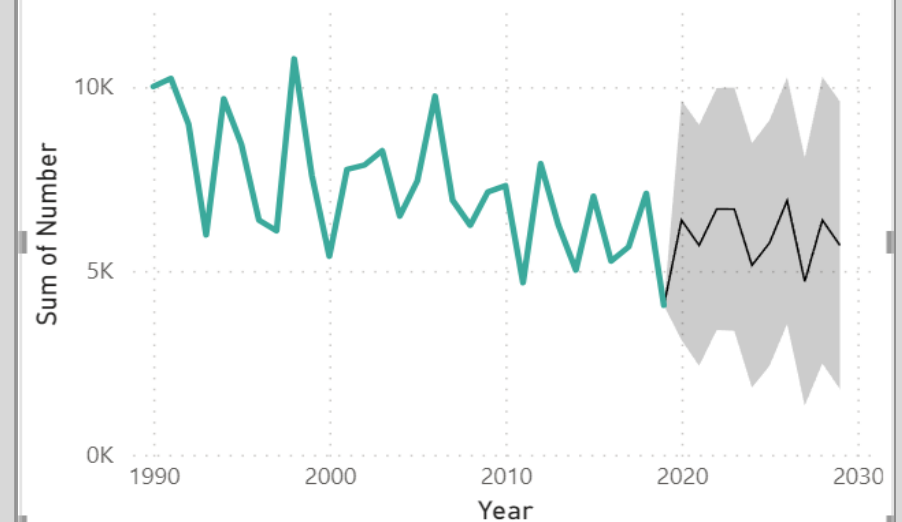


Average of precip by Year and conditions

conditions ● Rain, Overcast ● Snow, Overcast



Sum of Number by Year



PREDICTIVE ANALYSIS OF WILDFIRE



Logistic Regression

WITH K=17

[[37 32]
[26 61]]

	precision	recall	f1-score	support
0.0	0.59	0.54	0.56	69
1.0	0.66	0.70	0.68	87
accuracy			0.63	156
macro avg	0.62	0.62	0.62	156
weighted avg	0.63	0.63	0.63	156

SVM

	precision	recall	f1-score	support
0.0	0.65	0.29	0.40	69
1.0	0.61	0.87	0.72	87
accuracy			0.62	156
macro avg	0.63	0.58	0.56	156
weighted avg	0.62	0.62	0.58	156

[[20 49]
[11 76]]
Accuracy: 0.6153846153846154
Precision: 0.608
Recall: 0.8735632183908046
On Fire

Machine Learning Model

Random Forest

	precision	recall	f1-score	support
0.0	0.44	0.45	0.45	69
1.0	0.56	0.55	0.55	87
accuracy			0.51	156
macro avg	0.50	0.50	0.50	156
weighted avg	0.51	0.51	0.51	156

[[31 38]
[39 48]]
Accuracy: 0.5064102564102564
Precision: 0.5581395348837209
Recall: 0.5517241379310345
On Fire

Comparing Different models

LR: 0.391365 (0.150642)
KNN: 0.468401 (0.140563)
CART: 0.429676 (0.116502)
NB: 0.397549 (0.189845)
SVM: 0.480204 (0.234082)
MLP: 0.447097 (0.214353)
GradientBoost: 0.426094 (0.166582)
AdaBoost: 0.396908 (0.155590)
Bagging: 0.394759 (0.129567)
RandomForest: 0.392911 (0.141737)
ExtraTrees: 0.400566 (0.153712)

area 1
Month 0.3345
temp 0.0922
windspeed 0.0738
precip 0.0635
moisture -0.0331

Logistic Regression

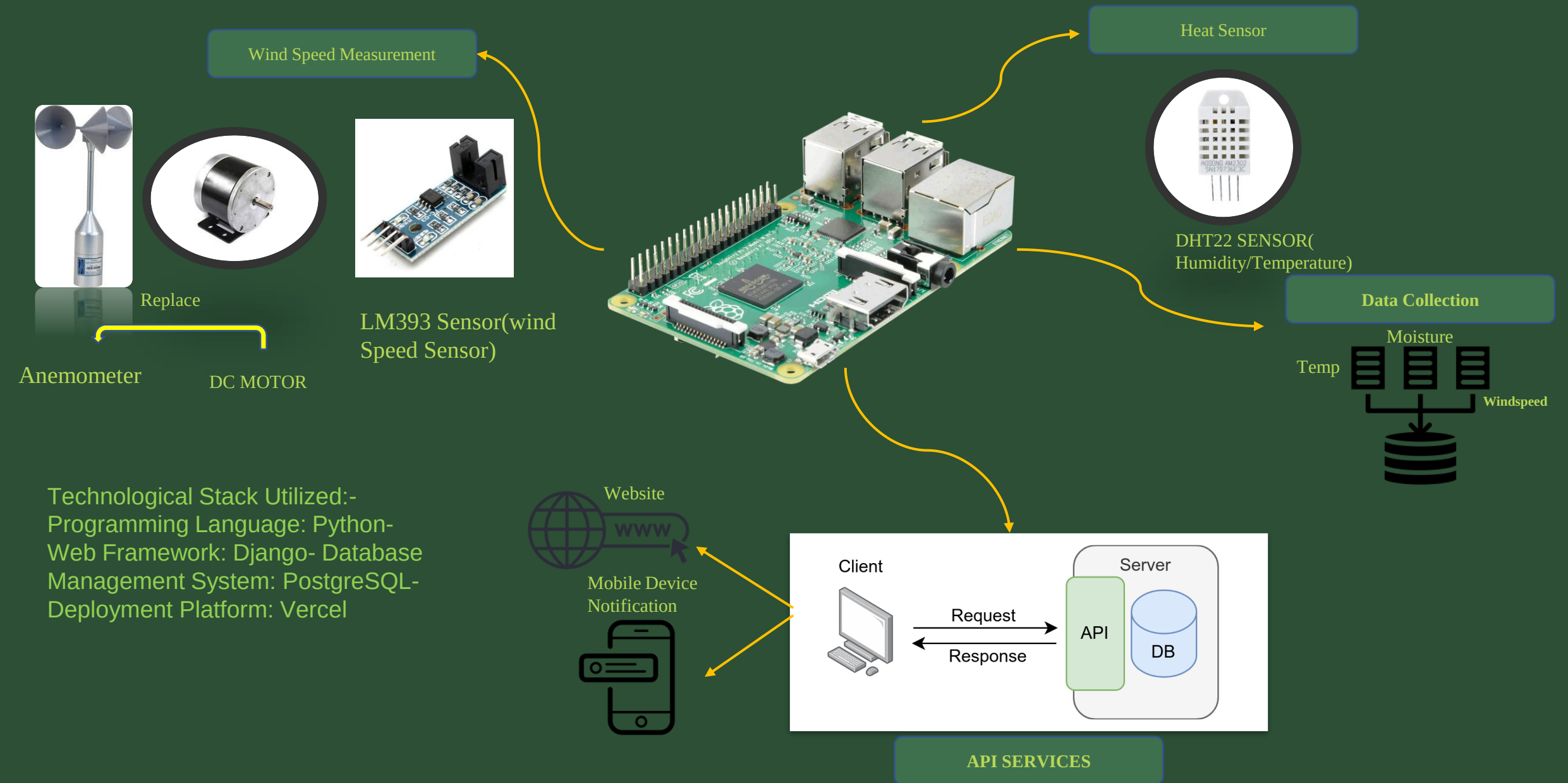
Accuracy: 0.48717948717948717
Precision: 0.5393258426966292
Recall: 0.5517241379310345
[[28 41]
[39 48]]

	precision	recall	f1-score	support
0.0	0.42	0.41	0.41	69
1.0	0.54	0.55	0.55	87
accuracy			0.49	156
macro avg	0.48	0.48	0.48	156
weighted avg	0.49	0.49	0.49	156

A full-page background image of a dense forest. The scene is misty, with soft sunlight rays (crepuscular rays) filtering through the canopy of trees. The trees are a mix of evergreens and deciduous trees with autumn-colored foliage in shades of yellow, orange, and red. The overall tone is serene and atmospheric.

Technological work

Wildfire prevention Tech prototype



Scan QR Code



Sensor Data

Temperature: 25.00 °C

Humidity: 62.96 %

Wind Speed: 23.50 km/h

The environmental conditions are currently safe.

Sensor Data

Temperature: 35.00 °C

Humidity: 62.96 %

Wind Speed: 23.50 km/h

Caution: The environmental conditions are conducive to the initiation and spread of wildfires.

Appendix

- Wildfire Prevention Measures for Urban Areas
 - Protocols can be adapted for urban environments (referred to as "**concrete jungles**").
- Data Enhancement Strategy
 - Valuable **dataset**; seeking **topography data, information on human life loss**, and detailed causes.
- User Experience Enhancement
 - Integration of additional **sensor data, including current readings**, for a more comprehensive dashboard.
- Future Opportunities
 - **Huge potential for using more ML models** to achieve finer and more accurate data.

Conclusion

- **Low-cost sensor prototype for proactive wildfire prevention**
- **Technological integration for safer coexistence with nature**
- **Power BI dashboard with dynamic features and key metrics**
- **Predictive modeling for strategic wildfire management**
- **Model accuracy at 48.72%, with precision and recall favoring wildfire occurrences; potential for enhanced prediction with strategic implications.**

Demo Links:

<https://drive.google.com/file/d/18k3XCVq1YaZJrPtWVTiEAVDbwf9zBCNO/view?usp=drivesdk>

https://drive.google.com/file/d/1JsUWZh8KZ7o0h7qitlIHVvcrIDX0P6BT/view?usp=drive_link



Thank You!